

When Memory Has to Remember Why

A Theory of Depth in Agent Memory

Abstract for a quick read

Modern AI agents have to carry information across very long tasks. The usual way they do this is to store everything as text and then, at each moment, pull back whatever looks most similar to the situation in front of them. This works for easy cases. It quietly breaks for hard ones. The reason is that the most important fact is often the one that does not look similar at all, because it sits several reasoning steps away and is written in different words. We call the distance between what truly matters and what merely looks similar the semantic and causal gap. This paper argues that similarity based memory has a depth ceiling, meaning its chance of recovering a chain of reasons falls away faster and faster as the chain gets longer. A memory that stores typed records joined by explicit links does not suffer this collapse, because it follows a link instead of guessing by resemblance. We show there is a clear depth at which the second kind of memory starts winning on both accuracy and cost at the same time, and we lay out exactly how to prove it and measure it. Brief is used as the working example of the second kind of memory.

1. The world the agent lives in

A modern agent works over horizons far longer than any single context window can hold. Across a long project, a large codebase, or a running sequence of product decisions, the agent keeps facing one question. Out of everything that has already happened, what matters right now. This is the memory problem, and the field has settled on one dominant answer. Store everything as text, turn it into embeddings, and at each step retrieve whatever is most similar to the current situation.

The trouble is that similarity is only a stand in for relevance, and it is a leaky one. The facts that matter most are often the facts that look least alike. A decision made early on quietly constrains an action taken much later, yet the two are phrased differently, live in different files, and sit far apart in embedding space. We give this idea a name. The gap between what is relevant and what merely looks relevant is the semantic and causal gap, and it is exactly where similarity retrieval fails without anyone noticing.

2. The chain of reasons

To make this precise, picture a chain of dependence of depth d ,

$$\mathcal{C} = (n_0 \rightarrow n_1 \rightarrow n_2 \rightarrow \cdots \rightarrow n_d).$$

The current task sits at the node n_0 . Acting correctly requires recovering every upstream link all the way back to n_d . A shallow chain has a small d . A deep chain, the kind that appears in real long running work, has a large d .

Definition 1 (Similarity decay). *Let $\rho \in (0, 1)$ be the rate at which resemblance to the query falls off as we move one causal step away from n_0 . The retrievability of the node at distance i is then*

$$s_i = s_0 \rho^i,$$

where $s_0 \in (0, 1]$ is the retrievability of the node closest to the query.

This is the heart of the matter. Each step away from where the agent is standing makes the next needed fact look a little less like the thing it is searching for.

3. Why similarity hits a ceiling

Suppose the system recovers each node independently with a chance equal to its retrievability. The probability that similarity retrieval recovers the whole chain is the product of the chances across every hop,

$$P_{\text{sim}} = \prod_{i=1}^d s_0 \rho^i = s_0^d \rho^{d(d+1)/2}.$$

Proposition 1 (Depth ceiling). *The full chain recovery probability of similarity retrieval falls off in the depth d at the rate $\rho^{d(d+1)/2}$, which is faster than any fixed exponential rate.*

The exponent $d(d+1)/2$ is the whole story in one symbol. For shallow chains the loss is mild and similarity is perfectly fine. For deep chains the recovery probability crashes toward zero, and the agent is reduced to guessing the very links that hold the task together.

4. The other kind of memory

Brief stores its memory in a different shape. Instead of a flat pool of embedded text, it holds typed records joined by explicit links. A decision, the constraint that the decision created, the feature that depends on that constraint, and the reasoning behind all of it, each connected by a labelled edge rather than by surface resemblance. Recovering a dependency is then no longer a similarity gamble. It is the act of following a pointer.

Let each link be recovered with probability q , close to one, failing only when a link was never written down or was malformed. Then the whole chain recovery probability is

$$P_{\text{struct}} = q^d.$$

This falls away only gently as the depth grows, because nothing here depends on whether two connected facts happen to look alike.

5. The moment the two cross

Place the two side by side. Their ratio is

$$\frac{P_{\text{struct}}}{P_{\text{sim}}} = \frac{q^d}{s_0^d \rho^{d(d+1)/2}} \xrightarrow{d \rightarrow \infty} \infty.$$

The two methods do not merely differ. The advantage of structured memory grows without bound as chains get deeper.

On pure recovery the structured store always looks better, so to be honest we must charge it for what it costs. Building and maintaining typed links takes real effort that a plain text pool does not pay. Let V be the value of solving a task correctly and let W be the extra overhead

the structured store pays per chain. Structured memory is the better choice when its net gain clears that overhead,

$$V(q^d - s_0^d \rho^{d(d+1)/2}) > W.$$

Write the left side as $g(d)$. Because the structured recovery stays high while the similarity recovery collapses, $g(d)$ rises as d grows. So there is a smallest depth at which the inequality first holds.

Proposition 2 (Crossover depth). *Define $g(d) = q^d - s_0^d \rho^{d(d+1)/2}$ and the overhead ratio $\tau = W/V$. Since g is increasing over the regime of interest, there exists a unique smallest integer depth d^* such that*

$$g(d) > \tau \quad \text{for all } d \geq d^*.$$

Below d^ the plain similarity store is good enough and cheaper. At and beyond d^* the structured store wins.*

Finding d^* , first by deriving it from ρ , q , s_0 , and τ , then by measuring it in real tasks, is the signature result of the paper.

6. Why the win is on cost as well as accuracy

The story does not stop at accuracy. Define the useful signal delivered to the model as the amount of context actually worth reading,

$$U = (\text{tokens injected}) \times (\text{retrieval precision}).$$

To make up for its low chance of catching each hop, similarity retrieval has to cast a wider net. It raises the number of retrieved items, pouring in more tokens while precision stays low. The structured store injects exactly the chain it followed, which is a small number of tokens at high precision. So once the depth passes d^* , the structured store wins on both axes at the same time. It is more often correct and it reads fewer tokens. That joint win is the strongest form of the claim and the one worth proving.

7. The thesis in one line

Similarity based memory has a depth ceiling. Its ability to recover a chain of reasons falls away faster than exponentially in the depth of the chain. Typed memory that follows explicit links escapes that ceiling, and there is a provable, measurable depth beyond which it wins on accuracy and on token cost together. We derive that depth and show it in practice using Brief.

8. What has to be proven, step by step

The storyline above turns directly into four pieces of work, each one a clean section of the eventual paper.

First, the decay model. Justify that resemblance to a node falls geometrically with causal distance, the claim $s_i = s_0 \rho^i$. This is checked by measuring embedding similarity along chains of dependence that are already known.

Second, the recovery bound. Derive P_{sim} and P_{struct} as products of the per step chances under stated assumptions, and be open about where the assumption of independence between hops starts to bend.

Third, the crossover. Solve for d^* , charge the structured store its honest overhead, and show that the win survives once that overhead is paid.

Fourth, the empirical match. On a set of tasks labelled by their depth, show that measured accuracy and measured token cost cross near the predicted d^* , with Brief serving as the structured arm.

The tension that carries the whole paper is simple and true. Shallow tasks hide the problem. Deep tasks expose it. Everything builds toward the depth at which the world tips over.
